

HS-HU601 — Economics for Engineers

Semester VI • 3L:0T:0P • 3 Credits

Module 4 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Depreciation Calculations

Assets lose value over time, which affects taxes and company value.

1.1 Depreciation Concepts

Depreciation: The systematic allocation of the cost of a tangible asset over its useful life to represent wear, tear, physical deterioration, or obsolescence.

Book Value (BV_t): The remaining value of an asset recorded in accounting books after subtracting accumulated depreciation from the initial cost.

1.2 Straight-Line (SL) Depreciation

The simplest method; allocates an equal amount of depreciation to each year of the asset's life.

Straight-Line Formulas

For initial cost C , salvage value S , and useful life n :

$$\text{Annual depreciation: } d_t = \frac{C - S}{n} \quad (1)$$

$$\text{Book value in year } t : BV_t = C - (t \times d_t) \quad (2)$$

1.3 Declining Balance (DB) & Double Declining Balance (DDB)

An accelerated method where depreciation is calculated as a fixed percentage of the remaining book value.

Declining Balance Formulas

The depreciation rate α is given by:

$$\alpha = \frac{\text{Factor}}{n} \quad (\text{For Double Declining Balance DDB, Factor} = 2) \quad (3)$$

Annual depreciation, book value, and accumulated depreciation calculations:

$$\text{Depreciation in year } t : d_t = \alpha \times BV_{t-1} \quad (4)$$

$$\text{Book value in year } t : BV_t = C \times (1 - \alpha)^t \quad (5)$$

Note: An asset cannot be depreciated below its salvage value. Once BV_t reaches S , depreciation stops.

2. Replacement Analysis

Replacement analysis determines when an existing asset should be replaced by a new one.

2.1 Economic Service Life (ESL)

The **Economic Service Life (ESL)** is the asset life k^* that minimizes the total equivalent annual cost. It is the optimum point that balances capital costs and rising operating costs.

Total Annual Cost Equation

The Total Annual Cost (ATC) of owning an asset for k years at interest rate i is:

$$ATC(k) = CR(k) + AEC_{O\&M}(k) \quad (6)$$

Where:

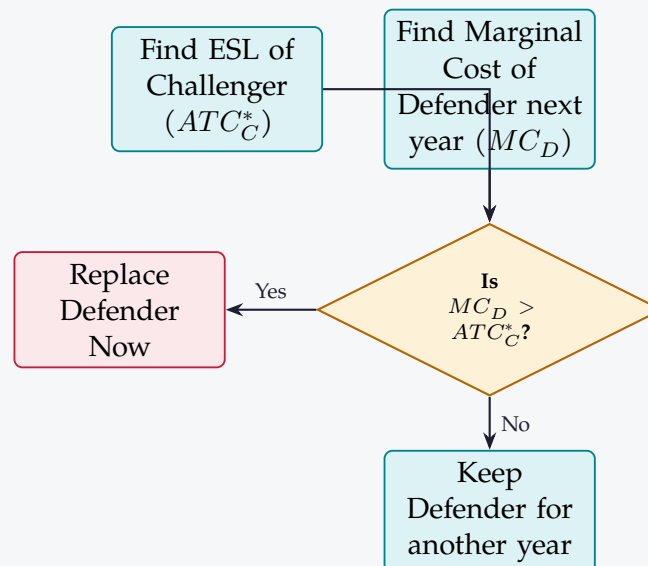
$$\text{Capital Recovery cost: } CR(k) = [C - S_k(P/F, i, k)] \times (A/P, i, k) \quad (7)$$

$$\text{Equivalent Annual O\&M: } AEC_{O\&M}(k) = \left[\sum_{t=1}^k O\&M_t \times (P/F, i, t) \right] \times (A/P, i, k) \quad (8)$$

2.2 Defender vs. Challenger Comparison

- **Defender (D)**: The existing asset currently in use.
- **Challenger (C)**: The best available replacement candidate.

Replacement Decision Process Flowchart



Marginal Cost of Defender next year:

$$MC_D(t+1) = (MV_t - MV_{t+1}) + (MV_t \times i) + O\&M_{t+1} \quad (9)$$

Where MV_t is current market value, MV_{t+1} is market value next year, and $O\&M_{t+1}$ is operating cost next year.

3. Basic Accounting for Engineers

Engineers must understand financial statements and cost allocation to make business decisions.

3.1 Financial Statements

1. **Balance Sheet:** Represents a snapshot of the firm's financial position at a specific point in time. It satisfies the fundamental accounting equation:

$$\text{Assets} = \text{Liabilities} + \text{Owner's Equity} \quad (10)$$

2. **Income Statement:** Summarizes revenues and expenses over a period of time (e.g., quarter or year) to calculate net income:

$$\text{Net Income} = \text{Revenues} - \text{Expenses} \quad (11)$$

3.2 Financial Ratios

- **Liquidity (Current Ratio):** $\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$
- **Profitability (Return on Equity):** $ROE = \frac{\text{Net Income}}{\text{Owner's Equity}}$
- **Leverage (Debt-to-Equity):** $\text{Debt-to-Equity} = \frac{\text{Total Liabilities}}{\text{Owner's Equity}}$

3.3 Cost Accounting and Allocation

- **Direct Costs:** Costs directly traceable to a product or service (e.g., direct materials, assembly labor).
- **Indirect Costs (Overhead):** Costs that cannot be easily traced to a specific product (e.g., building rent, supervisor salaries, factory electricity).
- **Indirect Cost Allocation:** Distributing overhead costs to products using allocation bases like direct labor hours, machine hours, or direct material costs.

4. Worked Examples

Example 1: Double Declining Balance (DDB)

Problem: A test instrument costs \$80,000, has a useful life of 5 years, and a salvage value of \$10,000. Calculate the depreciation amount and book value for Year 1 and Year 2 using the DDB method.

Solution: Identify parameters: Cost $C = \$80,000$, Life $n = 5$, Salvage value $S = \$10,000$. Calculate DDB rate α :

$$\alpha = \frac{2}{n} = \frac{2}{5} = 0.40 \quad (40\% \text{ per year})$$

- Year 1:

$$\text{Depreciation: } d_1 = \alpha \times BV_0 = 0.40 \times 80,000 = \$32,000$$

$$\text{Book Value: } BV_1 = C - d_1 = 80,000 - 32,000 = \$48,000$$

- Year 2:

$$\text{Depreciation: } d_2 = \alpha \times BV_1 = 0.40 \times 48,000 = \$19,200$$

$$\text{Book Value: } BV_2 = BV_1 - d_2 = 48,000 - 19,200 = \$28,800$$

Since $BV_2 = \$28,800 > S = \$10,000$, we do not need to adjust for the salvage value floor yet.

Answer: Year 1: Depreciation = \$32,000, Book Value = \$48,000. Year 2: Depreciation = \$19,200, Book Value = \$28,800.

Example 2: Replacement Analysis

Problem: An existing compressor (defender) has a current market value of \$5,000. If kept for another year, its market value will fall to \$3,500 and operating costs will be \$6,500. A new compressor (challenger) has a minimum annual cost of \$8,000. At an interest rate of 10%, should the company replace the compressor now?

Solution: Calculate the Marginal Cost of the defender (MC_D) for next year:

- Current market value $MV_t = \$5,000$, Market value next year $MV_{t+1} = \$3,500$.
- Interest rate $i = 0.10$.
- Operating and Maintenance cost $O\&M_{t+1} = \$6,500$.

$$MC_D = (MV_t - MV_{t+1}) + (MV_t \times i) + O\&M_{t+1}$$

$$MC_D = (5,000 - 3,500) + (5,000 \times 0.10) + 6,500$$

$$MC_D = 1,500 + 500 + 6,500 = \$8,500$$

Compare MC_D next year with the minimum annual cost of the challenger ($ATC_C^* = \$8,000$):

$$MC_D(\$8,500) > ATC_C^*(\$8,000) \quad (\text{Keeping defender is more expensive})$$

Since the marginal cost of keeping the defender exceeds the annual cost of the challenger, the defender should be replaced.

Answer: Replace now, as keeping the defender for another year costs \$8,500 compared to the challenger's annual cost of \$8,000.

Example 3: Indirect Cost Allocation

Problem: A workshop incurs factory overhead costs of \$12,000. It produces two products: Product A (requires 400 direct labor hours) and Product B (requires 800 direct labor hours). Allocate the overhead cost to both products using direct labor hours as the allocation base.

Solution: Identify parameters: Total Overhead = \$12,000, Total Labor Hours = 400 +

800 = 1,200 hours. Calculate the overhead allocation rate:

$$\text{Allocation Rate} = \frac{\text{Total Overhead}}{\text{Total Labor Hours}} = \frac{12,000}{1,200} = \$10 \text{ per labor hour}$$

Allocate overhead to each product:

- **Product A:**

$$\text{Allocated Overhead} = 400 \text{ hours} \times \$10/\text{hour} = \$4,000$$

- **Product B:**

$$\text{Allocated Overhead} = 800 \text{ hours} \times \$10/\text{hour} = \$8,000$$

Verification: $\$4,000 + \$8,000 = \$12,000$ (total overhead allocated).

Answer: Product A is allocated **\$4,000** and **Product B** is allocated **\$8,000**.

Quick Revision — Module IV

Depreciation, Replacement & Accounting | HS-HU601

Key Formulas at a Glance

Straight-Line Dep.	$d_t = \frac{C - S}{n} \quad (BV_t = C - t \times d_t)$
Declining Balance Dep.	$d_t = \alpha \times BV_{t-1} \quad (BV_t = C \times (1 - \alpha)^t)$
DDB Rate	$\alpha = \frac{2}{n}$
Capital Recovery	$CR(k) = [C - S_k(P/F, i, k)] \times (A/P, i, k)$
Defender Marginal Cost	$MC_D(t+1) = (MV_t - MV_{t+1}) + (MV_t \times i) + O\&M_{t+1}$
Accounting Equation	Assets = Liabilities + Owner's Equity
Current Ratio	Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$

One-Line Definitions

- **Depreciation:** Allocation of an asset's cost over its life to account for wear and tear.
- **Book Value:** Cost of an asset minus accumulated depreciation.
- **Economic Service Life:** The asset life that minimizes total equivalent annual cost.
- **Defender:** An existing asset currently owned and operating.
- **Challenger:** The best replacement candidate available.
- **Balance Sheet:** Statement showing assets, liabilities, and equity at a specific point in time.
- **Income Statement:** Statement showing revenues, expenses, and net profit over a period.
- **Direct Cost:** Expense directly traceable to a product (e.g., labor, raw materials).
- **Indirect Cost:** Expense that cannot be easily traced to a product (e.g., rent, utilities).

Frequently Asked / Viva Questions

Q1. What is the difference between physical deterioration and obsolescence?

Physical deterioration is wear and tear due to usage or environmental factors. Obsolescence is the loss of value due to technological progress (newer, better alternatives).

Q2. Explain the Straight-Line depreciation method.

It allocates an equal amount of depreciation to each year of the asset's useful life:

Frequently Asked / Viva Questions (contd.)

$$d_t = (C - S)/n.$$

Q3. What is the salvage value floor rule in Declining Balance methods?

An asset cannot be depreciated below its salvage value. Once the book value (BV_t) drops to the salvage value (S), depreciation must stop.

Q4. Define the Economic Service Life (ESL) of an asset.

The period of ownership that minimizes the sum of equivalent annual capital recovery costs and annual operating and maintenance costs.

Q5. What costs are included in the Capital Recovery cost (CR) of an asset?

The annualized cost of depreciation (loss in asset value) and the interest cost on the capital invested in the asset.

Q6. How is the defender's marginal cost for next year calculated?

By adding the projected loss in market value, the interest on its current market value, and the operating and maintenance costs for next year.

Q7. State the decision rule in Defender vs. Challenger replacement analysis.

If the marginal cost of keeping the defender for one more year is greater than the minimum annual cost of the challenger, replace the defender immediately.

Q8. State the fundamental accounting equation.

Assets = Liabilities + Owner's Equity. It forms the basis of the Balance Sheet.

Q9. What is the difference between direct and indirect costs in manufacturing?

Direct costs can be traced directly to a product (e.g., material). Indirect costs cannot be traced directly and must be allocated (e.g., factory rent).

Q10. What is the Current Ratio, and what does it measure?

Current Assets divided by Current Liabilities. It measures a firm's liquidity (ability to pay off short-term debts).

Q11. Explain the concept of overhead cost allocation.

The process of distributing indirect costs (like supervisor salaries or utilities) to different products using an allocation base like direct labor hours.

Q12. What is the Double Declining Balance (DDB) depreciation rate?

It is twice the straight-line rate: $\alpha = 2/n$. It represents accelerated depreciation in the early years of asset ownership.

Comparison of Depreciation Methods

Method	Depreciation Pattern	Calculation Base	Primary Use
Straight-Line	Constant annual depreciation	$(C - S)$ is constant	Standard accounting; simple financial reporting.
Declining Balance	Accelerated; higher in early years	Remaining book value (BV_{t-1})	Assets that lose value quickly (vehicles, computers).
Double Declining	Maximum acceleration rate ($2/n$)	Remaining book value (BV_{t-1})	Tax reporting (maximizes early tax shields).
Sinking Fund	Increases over time	$(C - S)$ with interest	Special asset funds; rarely used in modern tax accounting.